BMC SWITCHGEAR AND INSTRUMENT TRANSFORMER

BORN TO DISCOVER



BMC
EXCELLENCE REDEFINED
Since 1986

TECHNOLOGY CENTRE

The Fragrance of Knowledge

"Vision"

Our vision is to get recognised as one of the best "Global Medium Voltage R&D House". While carrying out the in-house R&D with "confidentially" as our top most focus, we also like to offer our R&D services worldwide along with the product supply support to the "Switchgear Houses" and the "Electrical Research Institutes".

"Mission"

In the field of medium voltage Switchgear and Instrument Transformers,

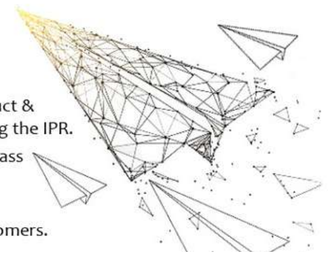
We,

.. Understand the future market needs in technology & product.

.. Carry out continuous innovation in product & process design and application protecting the IPR.

.. Support customer with in-house world class manufacturing & testing facility.

.. Subsidize the supply of finest quality of components & sub assembly to our customers.





The beginning of BMC

History is replete with giant companies that rose from humble beginnings. The great Indian engineer, Sir Vishveshwariya pronounced the immortal words, Industrialize or perish. In 1984 this stirred the minds of two Indian engineers Santanu Mitra & Subodh Banerjee from Kolkata. Later on they embarked on a journey of excellence in the Industry along with their team of directors Rajiv Bhargava, Siddhartha Mitra and Girish Magre.

USPs of BMC.....

- Patented design of switchgears & mechanisms.
- Pioneers in mechanism design in India.
- Pioneers in Epoxy cast Vacuum Interrupter (Embedded poles) for vacuum contactors worldwide.
- Unique compact design of MV Vacuum contactors.
- Over 25 patents/design registrations on Switchgears/parts of Switchgears.
- Unique low cost CSS design for windmills in India.
- First in India to offer technology of VCB /VI design to Indian manufacturers.
- First in Asia to offer technology of VCB /VI design to European & overseas companies.
- First in Indian SME sector to get accredited as "R&D centre" in switchgear field by DSIR (India).
- Represent as "Technical /Marketing Consultants" for High Power lab in Korea, Vacuum furnace manufacturer from Germany & component manufacturers for VI (Single window solution on Vacuum Switchgears)
- Experience to design & sell Retrofit VCBs of over 25 versions
- Experience of manufacturing custom built/own designed Epoxy Cast Dry Type Instrument Transformers (CTs/PTs) both for Indoor & Outdoor application
- Experience of manufacturing custom built/own designed Epoxy Cast Switchgear Parts

















MV Switchgears Components

- Mechanisms for 12, 24 & 36 kV VCBs
- Support insulators
- Drive insulators
- Moulded Spouts upto 5000 Amps
- Epoxy housing three in one
- Epoxy housing single
- Molded Vacuum Interrupters Indoor
- Molded Vacuum Interrupters Outdoor

MV Instrument Transformer

Epoxy Cast

- Indoor CTs upto 44 kA
- Indoor PTs
- Outdoor Dry Type CTs
- Outdoor Dry Type PTs

MV Vacuum Switchgears

- Retrofit Vacuum Circuit Breaker (VCB) Vertical & Horizontal Isolation
- Vacuum Contactors (VC) & Vacuum Contactor Panels (VCP) upto 12kV
- Retrofit with VC & Fuses on a trolley
- Modular Indoor VCB & VC Panel
- Outdoor VCB (Metal Clad & Porcelain Clad)
- VCB with Magnetic Actuator Drive (Indoor and Outdoor)
- Vacuum RMU
- Unitised Substations/Compact Substations
- MV Vacuum Switchgear
- Vacuum Interrupter
- MV Instrument Transformer
- MV Switch Board Panels
- Soft Starters



USPs of CSS

CSS accommodates VCB, CT, PT, CRP, Bus Duct etc. Hence compact in size therefore space required for Panel is less than a half w.rt

- Conventional outdoor arrangements
- Easy & faster in Erection & Commissioning
- Eco-friendly, no Oil based CTs, PTs; no gas filled CBs
- Maintenance Friendly
- Customized Design For Wind Mill Substation
- Minimal Foundation Work
- Low Life Cycle Cost
- Adoptable to various types of Transformers through Bus Ducts
- Safer
- Easy to Transport
- Vandalism Proof
- Typical Reference Standards: IS:14786:2 000, IEC 62271-100 & 200 (2012), IS:3156-(Part-1):2007, IS2705(Part:1):2007

BMC's

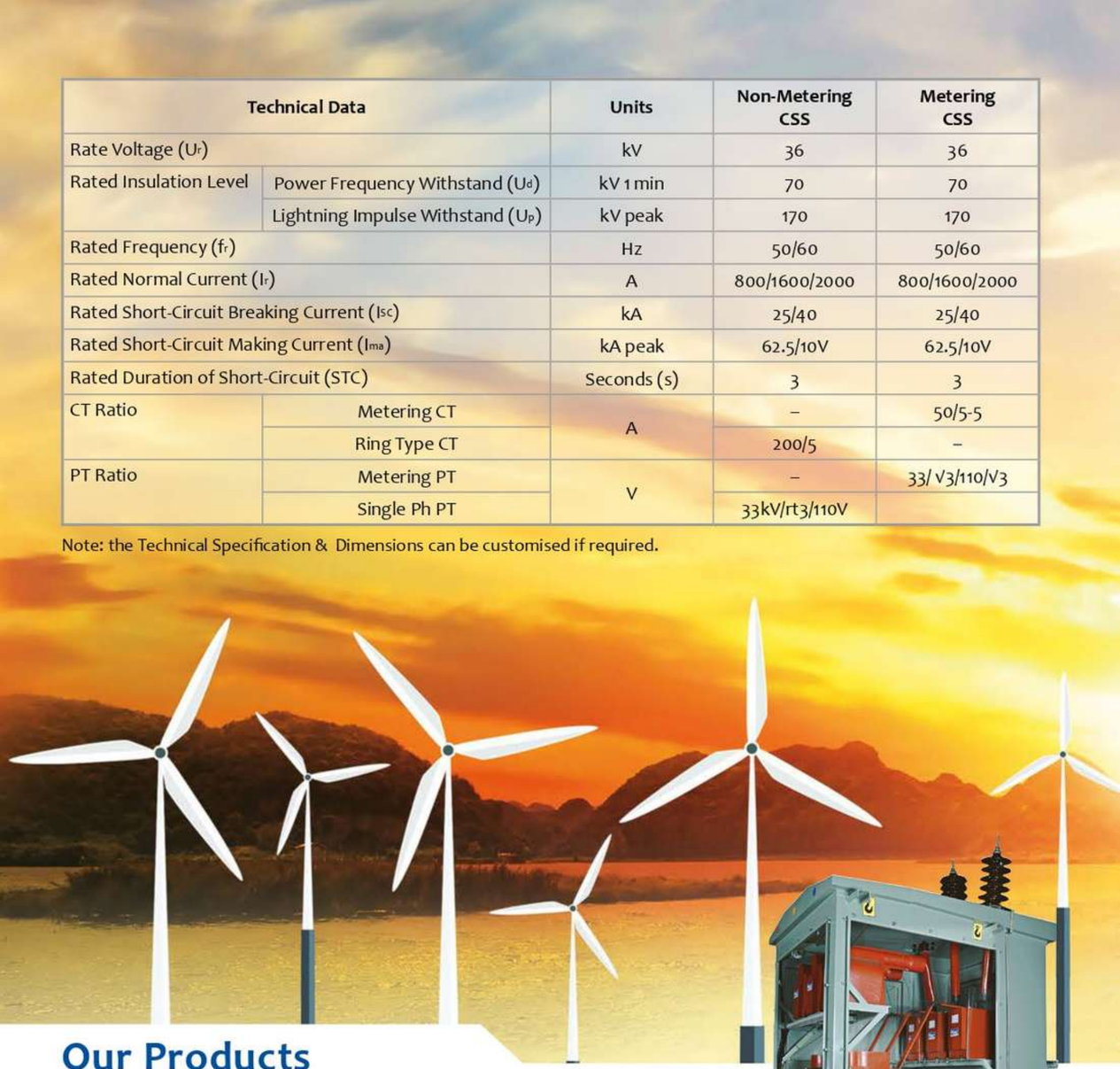
Compact Sub Station

Technical Data		Units	Non-Metering CSS	Metering CSS
Rate Voltage (U_r)		kV	36	36
Rated Insulation Level	Power Frequency Withstand (U_d)	kV 1 min	70	70
	Lightning Impulse Withstand (U_p)	kV peak	170	170
Rated Frequency (f_r)		Hz	50/60	50/60
Rated Normal Current (I_r)		A	800/1600/2000	800/1600/2000
Rated Short-Circuit Breaking Current (I_{sc})		kA	25/40	25/40
Rated Short-Circuit Making Current (I_{ma})		kA peak	62.5/10V	62.5/10V
Rated Duration of Short-Circuit (STC)		Seconds (s)	3	3
CT Ratio	Metering CT	A	–	50/5-5
	Ring Type CT		200/5	–
PT Ratio	Metering PT	V	–	33/ √3/110/√3
	Single Ph PT		33kV/rt3/110V	

Note: the Technical Specification & Dimensions can be customised if required.



CSS Accommodates VCB, CT, PT, CRP, Bus Duct



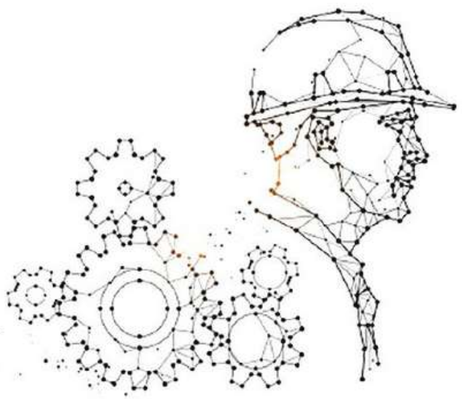
Our Products

Typical Products for Windmill Applications

- Compact Substations (CSS) upto 36kV
- Outdoor VCB (Metal Clad & Porcelain Clad) upto 36kV, 40kA
- Control Relay Panel (CRP)
- Vacuum Contactor (VC) & Vacuum Contactor Panels (VCP) upto 12kV
- Indoor Resin Cast CTs upto 12kV
- Indoor Resin Cast PTs
- Outdoor Dry Type Resin Cast CTs
- Outdoor Dry Type Resin Cast PTs
- LT Vacuum Contactor upto 1250A
- Ring Main Unit
- Load Break Switch
- Metering Cubical







Manufacturing

BMC, apart from the design and technology services is equipped with the manufacturing and testing facilities for MV Switchgear Components, MV Switchgear, MV Instrument Transformers, MV & LT Vacuum Contactors These facilities are located at Kolkata & Aurangabad in India.







Advanced Partner
Panel Builder
Accredited by Schneider Electric
Life Is On Schneider Electric

Manufacturer of MV Switchgear panels as core component advanced partner panel builder with Schneider Electric VCB up to 36kV system voltage.



36 kV SWITCHGEAR PANEL

Advanced Panel Builder And Global Supplier OEM To Schneider Electric For CT and PT



12 kV SWITCHGEAR PANEL



12kV SCHEINDER ELECTRIC VCB



36kV SCHEINDER ELECTRIC VCB





BMC hub at Kolkata

Manufacturing Facilities :

- Epoxy Casting Plant (Vacuum Casting & APG)
- Coil Winding & Assembly for CTs & PTs
- CRP, VCB & VC Assembly
- Mechanism Assembly
- Plating Shop
- Machine & Forging Shop
- CNC Fabrication (Tie-Up)
- Powder Coating (Tie-Up)
- Automatic Busbar Punching & Bending

USPs of BMC

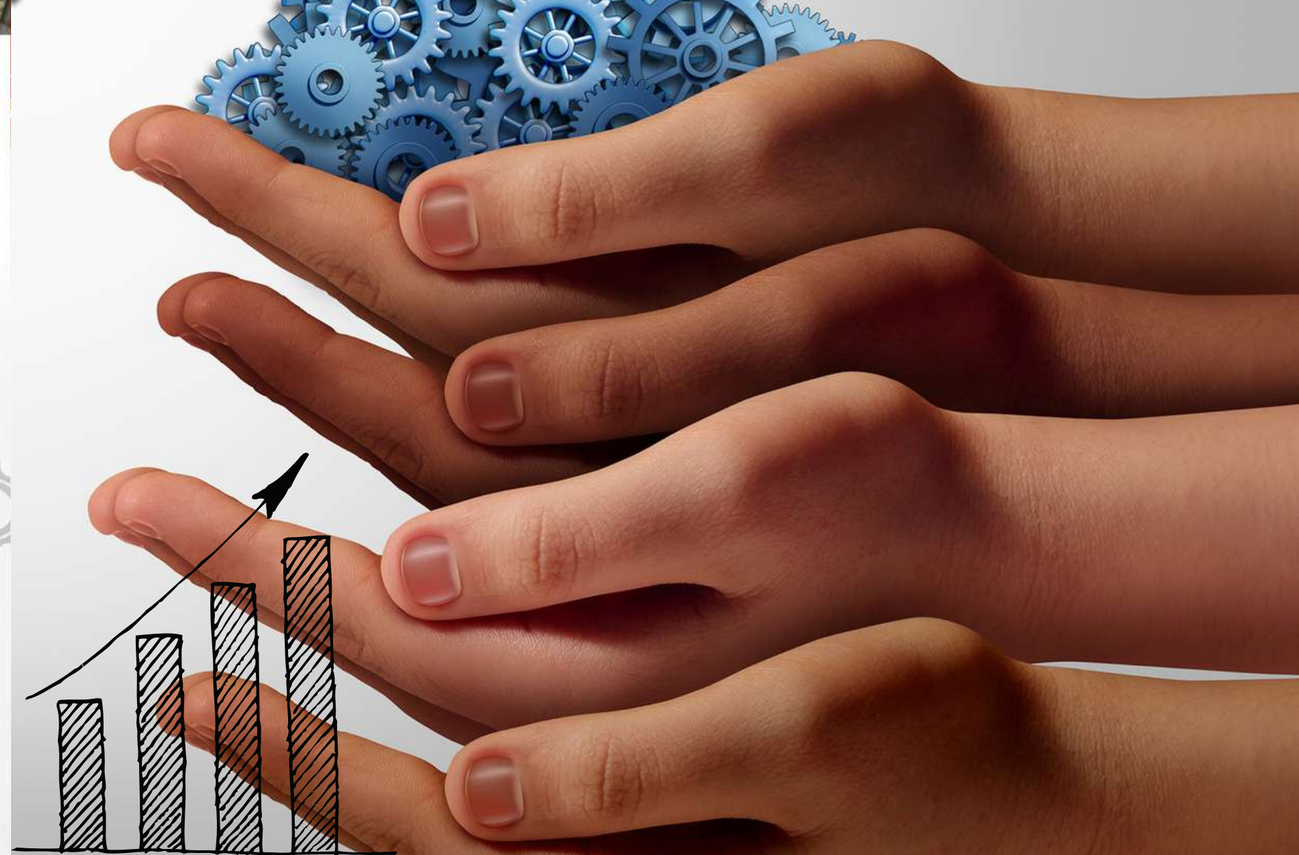
With Over Four Decades Of Experience In Power Distribution.

Ready To Move Skilled Engineers And Technicians With Hands-On Knowledge.

Cent Percent Problem Rectification On War Footing Basis.

Customized Economic Solution As Per Equipment Health, Installation Site Condition, Operator Suitability And Local Environment Condition.

Solution For Any And All Make Of Equipment From India And Abroad.



Services Offered By BMC

DIGITISATION FOR SUSTAINABILITY, HEALTH MONITORING AND DATA ANALYSIS OF COMPLETE POWER SYSTEMS.

COMPLETE ROUTINE TESTING AND DIAGNOSIS OF SUBSTATION EQUIPMENT AT SITE.

REPAIR, REPLACEMENT AND RETROFITTING OF COMPLETE SUBSTATION EQUIPMENT.

TAILOR MADE COMPONENTS FOR IMPORTED, CRITICAL AND OBSOLETE SUBSTATION EQUIPMENT.





APG-our specialty: The Automatic Pressure Gelation technique enables achievement of increased output as well as superior quality.
The advantages are as below-

- Low cycle times.
- Rapid production irrespective of size or complexity of shape.
- Reproducible dimensions within close tolerance.
- Cleaner process with reduced wastages.
- Reduction in rejection percentage.
- Improved mechanical properties and better surface finish

The hydraulic operated presses used make the moulding process accurate, fast, clean and hazard free. The Process Control parameters are accurately maintained with the help of advanced instrumentation
and PLC control.









A new age **research** and **design** organization

BMC ELECTOPLAST (I) Pvt. Ltd. The name BMC is synonymous with optimal solution in A thorough knowledge of electrical protection systems, designs of latest switching techniques and a consistent quality control system, assure the clients of durable and cost effective products and services for every imaginable purpose in switchgears.

BMC is one of the pioneer in using the latest software tools such as CFD, Thermal Imaging, FEA etc., for designing their product.

Success not by Accident but by **skill** and the **sense**

BMC ELECTROPLAST is geared to undertake turnkey projects based on a formal process developed inhouse. This process has inbuilt checks and balances to ensure that a client receives products and services of the highest quality. Feed back mechanisms provide management to have control over projects and greatly minimize chances of project overruns.











Testing Facilities:

- CT & PT Test setup
- PD Test setup
- Impulse Testing
- VCB Analyzer
- Mechanism Endurance Test Jig
- HV & MV Drop Test
- Temp rise Test
- Portable Vacuum Checker for Vacuum Interrupters
- Mechanical Testing facilities



Our Strengths

- We are an ISO 9001:2008 certified company.
- The Repetitive works clients have awarded us over the years.
- Our experience in working with most of the leading Switchgear Manufacture like Areva, Schnieders, Crompton Greaves, L&T, BHEL, Easun, Reyrolle, Vijai Electricals, GE, EATON of the Country.
- 100% track record of Completed Projects.
- Our Commitment to Quality, Speed and service.
- A well structured service back up team.

- Our International Presence.

  ...Global Switchgear Players like BATEL (Turkey), Tamco (Malaysia), Johnson, Philips (Dubai), KERI, ABM (Ukraine), Pommier - Group Cahores (France), Ormazabel (Spain) etc.
- Patent registration on many exclusive Product designs viz. Vacuum Interrupters, Vacuum Contactors, Dry type outdoor CTs, Mechanism.
- Type Testing of Product designs in KEMA, KERI, CPRI, ICMET International Labs.

Aurangabad the **Technology Centre** of **BMC**

A momentous decision was made in 2004, to further expand activities to Aurangabad, an industrial epicentre. Mr. Girish Magre, the enterprising electrical engineer and a global traveller, spearheaded the move. Six years later, the second plant at Aurangabad is designed considering the Eco-friendly green ambience, safe working stations and energy saving building architecture. The new State of Art manufacturing facility at Aurangabad- India has begun to be recognised as a Technology Centre, attracting many overseas Switchgear Houses. Enquiries to align the R&D work with product supply support had started pouring in.



Teamwork, safety, health environment:

At BMC Team work is one of the top most agenda, all the projects are discussed regularly at various stages. The active involvement of the team is ensured during the discussions. Though there are work freedom at all levels, the major decisions are taken collectively after principally reaching a consensus.

BMC ensures the health & safety of the entire stakeholders.



A societal responsibility:

We thus ensure contribution to that society as a responsible corporate citizens.





Our Partners

We are now representing the following companies : BATEL- Turkey, Sirui-China, PVA Tepla-Germany, etc.

BATEL elektromekanik

Sirui®

PVA TePla

★ Our partners subject to change



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Since 1986

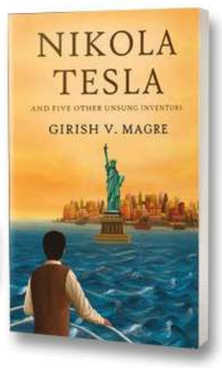
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Our contribution to Technology through a Technical Literature....



Available on amazon both hard and soft version...

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